Tobias Mwangi-Hartley

Data Scientist | Machine Learning, Forecasting and Experimentation
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PROFESSIONAL SUMMARY

Data scientist with experience developing forecasting, churn propensity and experimentation solutions. Built scheduled and batch-scored machine-learning workflows used in business operations, including a six-week recipe-demand forecast that improved WAPE from 23% to 14%. Hands-on with Python, SQL, BigQuery and Vertex AI Workbench, with experience in feature engineering, model evaluation and automated testing. Seeking to grow into the production engineering side of machine learning.

TECHNICAL SKILLS

Programming: Python, SQL, R
Machine learning and analytics: LightGBM, XGBoost, scikit-learn, pandas, statsmodels, SHAP, feature engineering, model evaluation, forecasting, experimentation
Tools: BigQuery, Vertex AI Workbench, dbt, Looker, Streamlit, Docker, pytest, R Markdown

PROFESSIONAL EXPERIENCE

Data Scientist | Thistlebox | 2022-Present

- Built a weekly recipe-demand forecasting workflow for six weeks ahead by recipe and region, using LightGBM with calendar, price and menu-position features.
- Improved forecast WAPE from 23% to 14%; the work was associated with an estimated 9% reduction in food waste. The scheduled Vertex AI Workbench notebook writes forecasts to BigQuery for operations reporting in Looker.
- Developed a weekly XGBoost churn-propensity scoring workflow in BigQuery and used top-decile scores to target retention offers. A 10% holdout validation found churn was down 2.1 percentage points in the treated group.
- Built a Streamlit calculator for planning and analysing sequential A/B tests, used by approximately 25 people in product and growth. Wrote the app and Dockerfile; the data platform team handles Cloud Run hosting and deployment.
- Since 2025, have written pytest unit tests for forecasting feature code, reaching approximately 60% coverage.
- Delivered the internal training “Stats for PMs” three times.

Junior Data Analyst | Firth Energy Retail | 2020-2022

- Segmented customers and analysed tariffs using R and SQL.
- Automated monthly regulator reports with R Markdown, saving the team approximately two days each month.

SELECTED PROJECTS

Image Classification Course Project | 2025

- Fine-tuned a small CNN in PyTorch on a public food-images dataset as part of an online deep-learning course, achieving approximately 80% accuracy. Learning project using PyTorch, torchvision and Google Colab.

EDUCATION

MSc in Statistics
Dissertation: Bayesian hierarchical models